Same grading rubric as Group A.

CS556-B Fall 2023 Midterm

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes.

Read carefully before answering! Write your answers on the white sheets given to you. Show all your work, not only the final answer.

Question 1 *(5 points)* If A and B are square and invertible, then AB is invertible, then $(AB)^{-1} = A^{-1}B^{-1}$.Explain.

False

$(AB)^{-1} = B^{-1}A^{-1}$

And the latter is not equal to $A^{-1}B^{-1}$.

Question 2 *(5 points)* If A is a 3-by-3 matrix and the equation $Ax = b$ has a unique solution, then A is invertible? Explain.

True. If the system has a unique solution, then there is no free variable, so A is invertible.

Question 3 *(5 points)* If a system of linear equations has two different solutions, it must have infinitely many solutions. True or False? Explain.

True. Since a system can have 0, 1 or ∞ solutions. If it has 2, then it has an infinite number of solutions.

Question 4 *(5 points)* If B, C, D are three matrices and $BC = BD$, then $C = D$. True or False?

False. $B(C-D)=0$ for C different from D.

Question 5 *(5 points)* The columns of an invertible m-by-m matrix form a basis for $\mathcal{R}^m$. True or False? Explain.

The columns of an invertible m-by-m matrix form a set of m independent vectors, which form a basis for R^m. True.

Question 6 *(25 points)* Let $x_1 = \begin{bmatrix}2\\3\\2\end{bmatrix}$, $x_2 = \begin{bmatrix}1\\-1\\2\end{bmatrix}$, and $v = \begin{bmatrix}a\\b\\c\end{bmatrix}$.

a) Find a vector x_3 such that $\{x_1, x_2, x_3\}$ is a linearly independent set of vectors.

b) Find the determinant of the 3-by-3 matrix composed of columns x_1, x_2, x_3.

c) Find the rank of this matrix.

d) Find the inverse of this matrix.

e) Find a condition that a, b, c must satisfy for the set of the vectors $\{x_1, x_2, v\}$ to be linearly dependent.

a) In order for the set $\{x_1, x_2, x_3\}$ is linearly independent, the determinant of matrix $A = [x_1\ x_2\ x_3]$ should be different from zero.

Let $x_3 = \begin{bmatrix}3\\2\\5\end{bmatrix} \Rightarrow A = \begin{bmatrix}2 & 1 & 3\\3 & -1 & 2\\2 & 2 & 5\end{bmatrix}$

$$\det(A) = 2\begin{vmatrix}-1 & 2\\2 & 5\end{vmatrix} - 1\begin{vmatrix}3 & 2\\2 & 5\end{vmatrix} + 3\begin{vmatrix}3 & -1\\2 & 2\end{vmatrix}$$

$$= -18 - 11 + 24 = -5 \neq 0$$

Hence, the set of vectors is linearly independent.

b) $\det(A) = -5 \neq 0$, found on a.

c) $\text{rank}(A) = 3$, since the vectors are linearly independent

d) Using Gauss-Jordan

$$A^{-1} = \begin{bmatrix}9/5 & -1/5 & -1\\11/5 & -4/5 & -1\\-8/5 & 2/5 & 1\end{bmatrix}$$

e) In order for the set to be linearly dependent $\det(A) = 0$

$$A = \begin{bmatrix}2 & 1 & a\\3 & -1 & b\\2 & 2 & c\end{bmatrix} \Rightarrow \det(A) = a\begin{vmatrix}3 & -1\\2 & 2\end{vmatrix} - b\begin{vmatrix}2 & 1\\2 & 2\end{vmatrix} + c\begin{vmatrix}2 & 1\\3 & -1\end{vmatrix}$$

$$= 7a - 2b - 5c$$

$$\boxed{7a - 2b - 5c = 0}$$

Question 7 *(25 points)* Given matrix $A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$.

a) Find $N(A)$ (the nullspace of A).

b) What is the dimension of $N(A)$?

c) Determine 3 vectors that are in the nullspace of A.

d) Calculate the inner product of these 3 vectors with one of the rows of A.

a) $\begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \xrightarrow[R1/2]{R2=R2+R1} \begin{bmatrix} 1 & 2 & -1 & 0.5 \\ 0 & -1 & 5 & 4 \\ 3 & 7 & -8 & 6 \end{bmatrix} \xrightarrow{R3=R3-3R1} \begin{bmatrix} 1 & 2 & -1 & 0.5 \\ 0 & -1 & 5 & 4 \\ 0 & 1 & -5 & 4.5 \end{bmatrix}$

$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & -1 & 0.5 \\ 0 & 1 & -5 & 4.5 \\ 0 & -1 & 5 & 4 \end{bmatrix} \xrightarrow{R3=R3+R2} \begin{bmatrix} 1 & 2 & -1 & 0.5 \\ 0 & 1 & -5 & 4.5 \\ 0 & 0 & 0 & 17/2 \end{bmatrix} \xrightarrow{R3 * 2/17}$

$\begin{bmatrix} \textcircled{1} & 2 & -1 & 0.5 \\ 0 & \textcircled{1} & -5 & 4.5 \\ 0 & 0 & 0 & \textcircled{1} \end{bmatrix}$ $\begin{matrix} x & y & z & w \end{matrix}$

z is the free variable

$z = 1$

$w = 0$

$y = 5$

$x = -9$

$N(A) = \begin{bmatrix} -9 \\ 5 \\ 1 \\ 0 \end{bmatrix} s, \; s \in \mathbb{R}$

b) $\dim(N(A)) = 1$

c) 3 vectors: $\begin{bmatrix} -9 \\ 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -18 \\ 10 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -27 \\ 15 \\ 3 \\ 0 \end{bmatrix}$

d) All inner products are 0, since the nullspace is orthogonal with the rowspace.

Question 8 *(25 points)* Consider the following system of linear equation for some constant k:

$$x + 2y - z = -2$$
$$5x - 2y + z = 12$$
$$-3x + 6y - 3z = k$$

a) For which value(s) of k does this system have one or infinitely many solutions?

b) Find all solutions for each value of k.

a) $\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 5 & -2 & 1 & 12 \\ -3 & 6 & -3 & k \end{array}\right]$ $\begin{array}{l} R2=R2-5R1 \\ R3=R3+3R1 \end{array}$ $\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & -12 & 6 & 22 \\ 0 & 12 & -6 & k-6 \end{array}\right]$ $R3=R3+R2$ →

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & -12 & 6 & 22 \\ 0 & 0 & 0 & k+16 \end{array}\right]$ $R2/-12$ → $\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & 1 & -1/2 & -11/6 \\ 0 & 0 & 0 & k+16 \\ x & y & z & \end{array}\right]$

In order for the system to have infinitely many solutions:

$k+16 = 0 \Rightarrow k = -16$

b) - Xparticular

Set the free variable z to 0

$z = 0$

$y = -11/6$

$x = 5/3$

$X_{particular} = \begin{bmatrix} 5/3 \\ -11/6 \\ 0 \end{bmatrix}$

- Xnullspace

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \\ x & y & z & \end{array}\right]$

Set the free variable $z = 1$

$z = 1$

$y = 1/2$

$x = 0$

$X_{nullspace} = \begin{bmatrix} 0 \\ 1/2 \\ 1 \end{bmatrix} s, s \in \mathbb{R}$

$X = X_{particular} + X_{nullspace}$

$= \begin{bmatrix} 5/3 \\ -11/6 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/2 \\ 1 \end{bmatrix} s, \forall s \in \mathbb{R}$

Question 9 *(5 points)* [Bonus] Please take a few minutes to share your thoughts on the course so far. Please be thoughtful and frank. Thank you.

- What do you like about the course so far?
- What suggestions do you have to improve this course?
- Do you have any other comments?